APACHE HADOOP MAPREDUCE

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Aim:

To implement an Apache Hadoop MapReduce program using Python and mrjob to analyse web server access logs stored in HDFS and calculate the total bandwidth utilized by each client IP address.

Problem Statement:

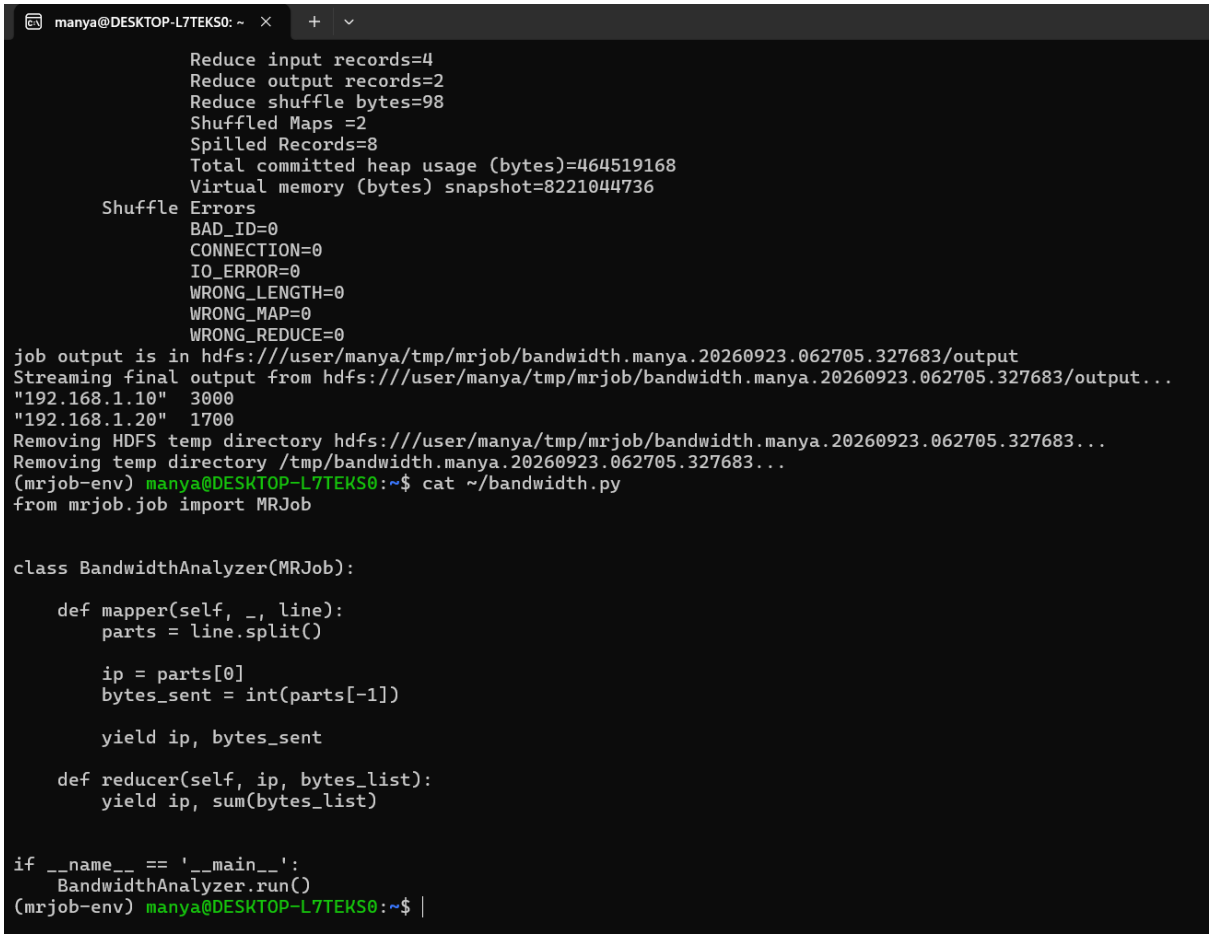
A web hosting provider stores raw web server access logs in the Hadoop Distributed File System (HDFS). The objective is to analyse these logs and determine the total number of bytes transferred for each client IP address.

The MapReduce program extracts the client IP address and response size from each log entry. The Map phase generates the IP address and corresponding bytes transferred, while the Reduce phase calculates the total bytes transferred for each IP address.

Code Implementation:

Python MapReduce Program

bandwidth.py



The program uses the Mapper to extract the client IP address and the number of bytes transferred. The Reducer groups the values by IP address and calculates their total bandwidth.

Execution Output:

The input web server log contains the following records:

192.168.1.10 - - [22/Sep/2026:10:00:01] "GET /index.html HTTP/1.1" 200 500

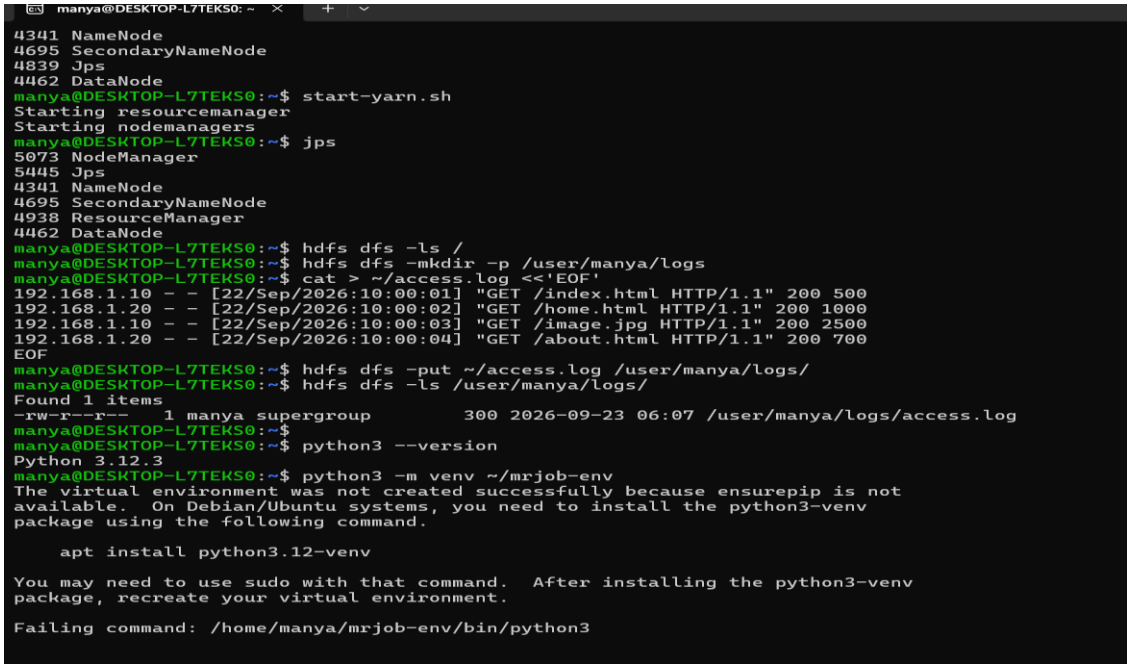
192.168.1.20 - - [22/Sep/2026:10:00:02] "GET /home.html HTTP/1.1" 200 1000

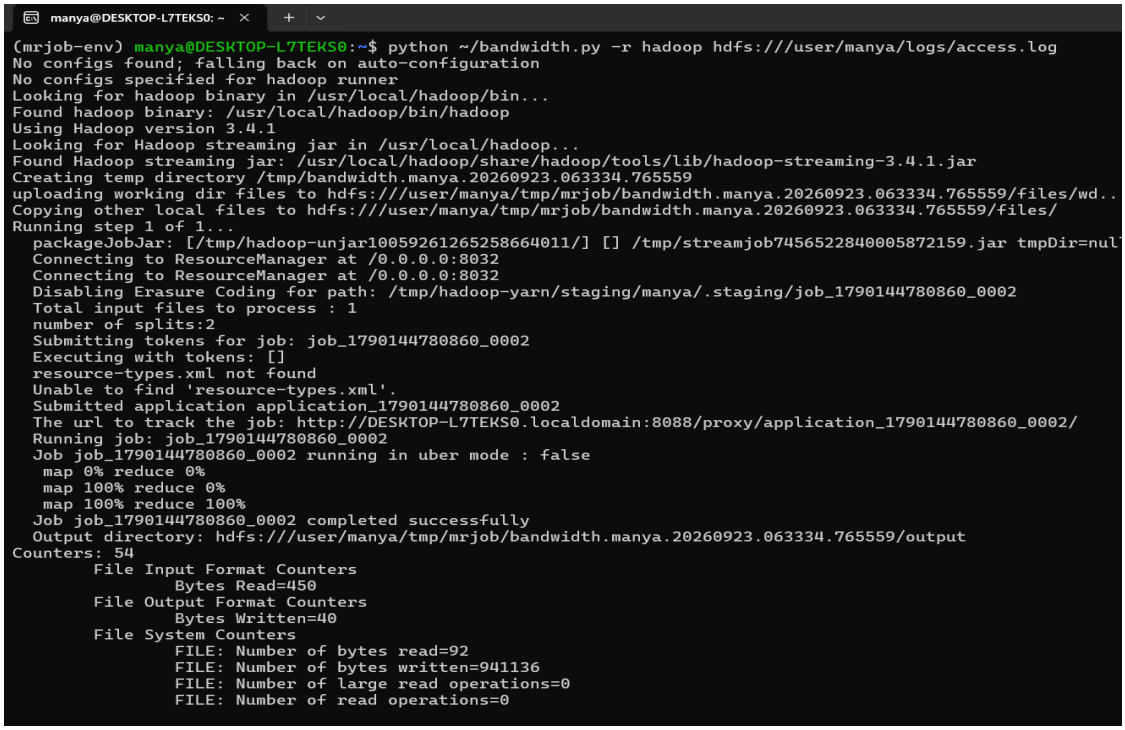
192.168.1.10 - - [22/Sep/2026:10:00:03] "GET /image.jpg HTTP/1.1" 200 2500

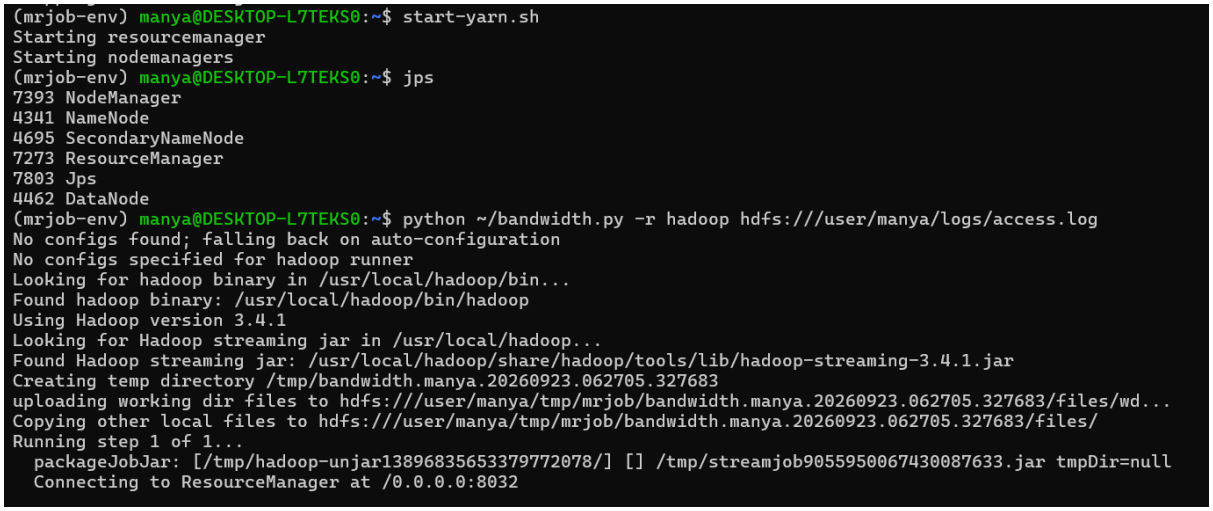
192.168.1.20 - - [22/Sep/2026:10:00:04] "GET /about.html HTTP/1.1" 200 700

The input file was stored in HDFS as:

/user/manya/logs/access.log







Brief Discussion:

The MapReduce program processes the access log using two stages. In the Map stage, each log entry is read and the client IP address and response size are extracted. In the Reduce stage, all byte values belonging to the same IP address are grouped and added together.

For IP address 192.168.1.10:

500 + 2500 = 3000 bytes

For IP address 192.168.1.20:

1000 + 700 = 1700 bytes

Result:

The Hadoop MapReduce job was executed successfully and produced the following result:

192.168.1.10 3000

192.168.1.20 1700

This gives the total bandwidth transferred by each client IP address.

Conclusion:

The Apache Hadoop MapReduce program was successfully implemented using Python and mrjob. The web server access log was stored and processed using HDFS and Hadoop MapReduce. The program successfully calculated the total bandwidth utilization for each client IP address.